4-21-26 (WEEK 9)

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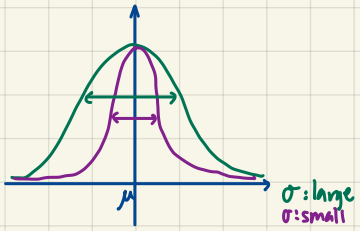
NORMAL DISTRIBUTION

$X \sim N(\mu, \sigma^2)$: $\mu \in \mathbb{R}$: mean, $\sigma \in \mathbb{R}$: std dev



PDF: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$, $f_X(x) > 0 \ \forall x \Rightarrow \forall a<b$, $\mathbb{P}(a \le X \le b) > 0$

Check: $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ ($\Rightarrow f_X$: valid PDF)

Proof

Change of variables: $z = \frac{x-\mu}{\sigma} \Rightarrow dx = \sigma dz$

$$\therefore I = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \sigma dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz$$

$$I^2 = \frac{1}{2\pi}\left(\int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz\right)\left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy\right) = \frac{1}{2\pi}\iint_{\mathbb{R}^2} e^{-\frac{z^2+y^2}{2}} dz\,dy$$

Switch to polar coordinates: $z = r\cos\theta$, $y = r\sin\theta$, $\begin{vmatrix} z_r & z_\theta \\ y_r & y_\theta \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$

$$\Rightarrow I^2 = \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\infty} e^{-\frac{r^2}{2}} r\,dr\,d\theta = \int_0^{\infty} e^{-\frac{r^2}{2}} r\,dr = \left[-e^{-\frac{r^2}{2}}\right]_0^{\infty} = 1$$

$\because I > 0$

$\therefore I = 1$ $\square$

DEFINITION

Standard Normal Distribution: $N(0,1)$

OBSERVE

If $X \sim N(\mu, \sigma)$, then $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$ ($z = \frac{x-\mu}{\sigma}$ is called the z-score)

Proof

$\mathbb{P}(Z \le z) = \mathbb{P}\left(\frac{X-\mu}{\sigma} \le z\right) = \mathbb{P}(X \le \mu + \sigma z) = \int_{-\infty}^{\mu+\sigma z} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$

Change of var: $t = \frac{x-\mu}{\sigma} \Rightarrow dx = \sigma dt$

$\therefore \mathbb{P}(Z \le z) = \int_{-\infty}^{z} \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}}_{\text{PDF for } N(0,1)} dt = \mathbb{P}(N(0,1) \le z)$

QUANTITIES

Expectation: For $Z \sim N(0,1)$, $\mathbb{E}[Z] = \int_{-\infty}^{\infty} t f_Z(t)\,dt = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} t e^{-\frac{t^2}{2}} dt = \frac{1}{\sqrt{2\pi}}\left[-e^{-\frac{t^2}{2}}\right]_{-\infty}^{\infty} = 0$

$\therefore$ For $X \sim N(\mu, \sigma)$, $Z = \frac{X-\mu}{\sigma} \sim N(0,1) \Rightarrow \mathbb{E}(X) = \mathbb{E}[\mu + \sigma Z] = \mu + \sigma\mathbb{E}[Z] = \mu$

Variance: For $Z \sim N(0,1)$, $\mathbb{E}[Z^2] = \int_{-\infty}^{\infty} t^2 f_Z(t)\,dt = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} t(te^{-\frac{t^2}{2}})\,dt \overset{\text{IBP}}{=} \cancel{\frac{1}{\sqrt{2\pi}}\left[te^{-\frac{t^2}{2}}\right]_{-\infty}^{\infty}}^{\,0} + \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt = \int_{-\infty}^{\infty} f_Z(t)\,dt = 1 \Rightarrow Var(Z) = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = 1 - 0^2 = 1$

$\therefore$ For $X \sim N(\mu, \sigma)$, $Z = \frac{X-\mu}{\sigma} \sim N(0,1) \Rightarrow \underbrace{Var(X) = \mathbb{E}[(X-\mu)^2]}_{\text{def of var}} = \mathbb{E}[(\sigma Z)^2] = \sigma^2\mathbb{E}[Z^2] = \sigma^2$

EXAMPLE

There are 2 envelopes, one with \$x, the other with \$2x. We don't know x.

Choose an envelope, look inside, then decide to switch or not.

What should you do to maximize your chance of getting more money?

Solution

Say we have the envelope with \$y.

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Case 1: $y=x$, other env $2y$ (50% chance)
Case 2: $y=2x$, other env $\frac{1}{2}y$ (50% chance)
$\Rightarrow$ If switch, $\mathbb{E}[\text{payout}] = 50\%(2y) + 50\%(\frac{1}{2}y) = \$\frac{5}{4}y$ (but still prob 50% switch is bigger env)

However, we CAN beat 50% chance of getting the bigger env

adjusted to $[0,\infty)$ money scale but realistic

Method: $Z \sim N(0,1)$, and sample from Z to get z. If $y<z$, switch. Else, keep

Case 1: $z<x \Rightarrow$ always keep (50% winning)
Case 2: $z>2x \Rightarrow$ always switch (50% winning)
Case 3: $x \leq z \leq 2x \Rightarrow$ always big env (100% winning)

By law of total prob,

$$\mathbb{P}(\text{win}) = \mathbb{P}(\text{win} \mid z<x)\,\mathbb{P}(z<x) + \mathbb{P}(\text{win} \mid x \leq z \leq 2x)\,\mathbb{P}(x \leq z \leq 2x) + \mathbb{P}(\text{win} \mid z>2x)\,\mathbb{P}(z>2x)$$
$$= 50\%\left[\mathbb{P}(z<x) + \mathbb{P}(x \leq z \leq 2x) + \mathbb{P}(z>x)\right] + 50\%\,\mathbb{P}(x \leq z \leq 2x) = 50\% + 50\%\,\underbrace{\mathbb{P}(x \leq z \leq 2x)}_{>0} > 50\%$$

CUMMULATIVE DISTRIBUTION FUNCTION

$F_Z(t) = \mathbb{P}(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}\,dt$

But $\nexists$ closed form antiderivative

$\therefore$ We define $\mathbb{P}(Z \leq z) =: \Phi(z)$ which is to be calculated numerically

REMARKS

① $\mathbb{P}(a \leq Z \leq b) = \mathbb{P}(Z \leq b) - \mathbb{P}(Z \leq a) = \Phi(b) - \Phi(a)$

② $\mathbb{P}(Z \geq a) = 1 - \mathbb{P}(Z \leq a) = 1 - \Phi(a)$

③ Symmetry: $\mathbb{P}(Z \leq z) \overset{\text{sym}}{=} \mathbb{P}(Z \geq -z) \overset{\text{comp}}{=} 1 - \mathbb{P}(Z \leq -z) = 1 - \Phi(-z)$

$\therefore$ It suffices to know $\Phi(z)$ $\forall z \geq 0$

④ If $X \sim N(\mu, \sigma^2)$, then $\mathbb{P}(X \leq x) = \mathbb{P}(Z \leq \frac{x-\mu}{\sigma}) = \Phi(\frac{x-\mu}{\sigma})$ "z-score": # std dev above mean

EXAMPLE

Say we're transmitting binary data $\in \{0,1\}$ over a noisy channel.
Signal received: Signal sent + $N(0, 0.3^2)$.

Decoding algo: If we receive x, $\begin{cases} x \geq 0.5 \rightarrow \text{interpret as } 1 \\ x < 0.5 \rightarrow \text{interpret as } 0 \end{cases}$

Signal sent:

Let Y = noise, $Y \sim N(0, 0.3^2)$, $\mathbb{P}(\text{mistake}) = ?$